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Real-Time Multimodal Detection of Fraud and Scam Calls Using Conversation Stage Modeling and Ensemble Learning

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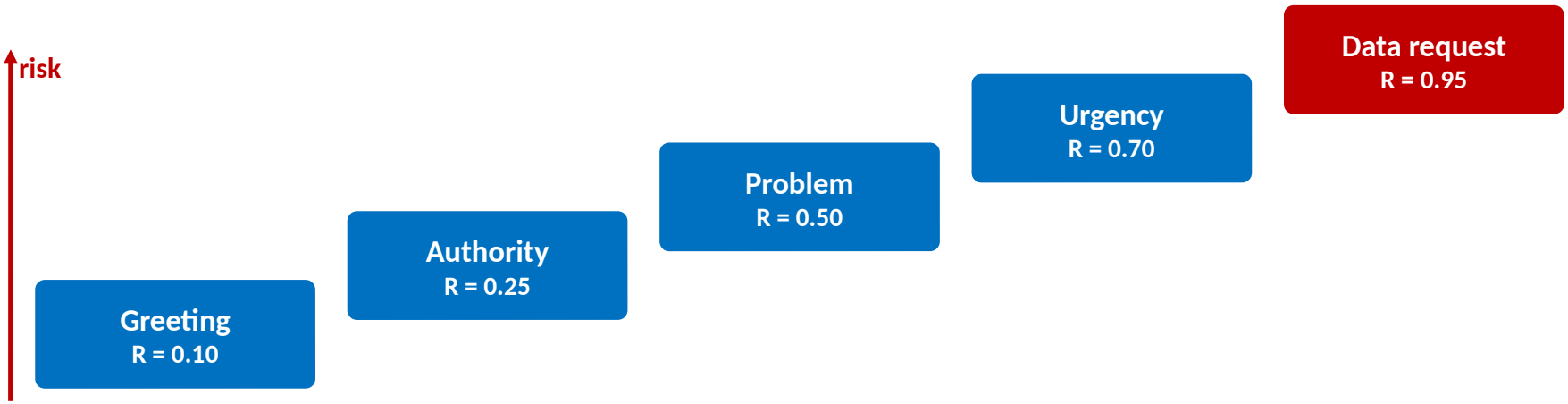
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github.com/CyberKnight-cmd/spam-call-detection

Problem and Key Idea

- **Social engineering:** the caller persuades; nothing is hacked
- **Damage happens during the call**, so the warning must too
- **Prior work:** blacklists [1], [2]; call records [3], [4]; transcripts after the call [5]



Scam calls follow a fixed script [6]: know the stage → know the danger

1	2	3	4
Conversation-stage modelling, live	**Four-model** ensemble	**Weighted fusion** + EMA smoothing	**Working open system** app + server + code

Framework: From Audio to Features



Step 1 · Cleaning each window

$$\Lambda_\ell = \log \frac{p(\mathbf{x}_\ell \mid \text{speech})}{p(\mathbf{x}_\ell \mid \text{noise})} > \tau$$

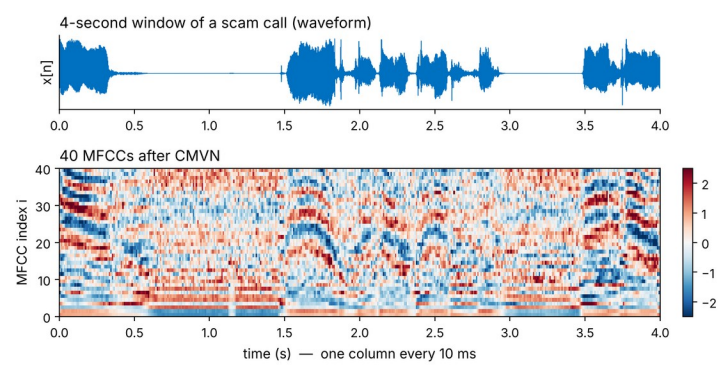
Voice activity [7]: 30 ms frames · skip window if speech < 1 s

$$E_t = \sqrt{\frac{1}{N}\sum_{n=0}^{N-1} x_t[n]^2}$$

$$\text{keep } t \quad \text{if} \quad E_t > 0.6\,\bar{E}$$

dominant speaker: runs ≥ 1.2 s

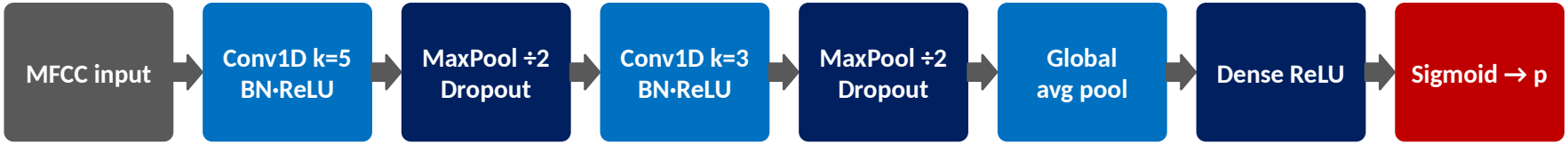
Step 2 · MFCC: FFT → Mel → log → DCT [8]



$$c_t(i) = \sum_{m=1}^{M} \log S_t(m) \cos\left[\frac{\pi i\,(m-\frac{1}{2})}{M}\right]$$

40 MFCC + Δ + ΔΔ [9] → input 370 × 120

Acoustic Models: Two 1-D CNNs



	Model 1 · Phoneme CNN	Model 3 · Repetition CNN
Input	370 × 120 (MFCC + Δ + ΔΔ)	4 segments × (470 × 40 MFCC)
Filters	64 (k = 5) → 128 (k = 3)	32 (k = 5) → 64 (k = 3)
Parameters	≈ 80.6 k	≈ 17.2 k
Learns	phoneme-scale sound (≈ 120 ms)	scripted, repeated phrases [15]
Output	scam probability	max of 4 segment probabilities

1-D convolution [10]

$$h_{t,\,f} = \mathrm{ReLU}\left(b_f + \sum_{k=0}^{K-1}\sum_{c=1}^{C} W_{f,\,k,\,c}\; x_{t+k,\,c}\right)$$

- Same weights W at every time step
- BatchNorm [11] · Dropout [12] · GAP [13]

Segment max · training [14]

$$p_{\mathrm{rep}} = \max_{j=1,\,\ldots,\,4} g_\phi(\mathrm{MFCC}_{40}(s_j))$$

$$\mathcal{L} = -\frac{1}{N}\sum_{i=1}^{N}[y_i \log p_i + (1-y_i)\log(1-p_i)]$$

Adam, η = 5×10^{-4} (phoneme) · 10^{-3} (repetition)

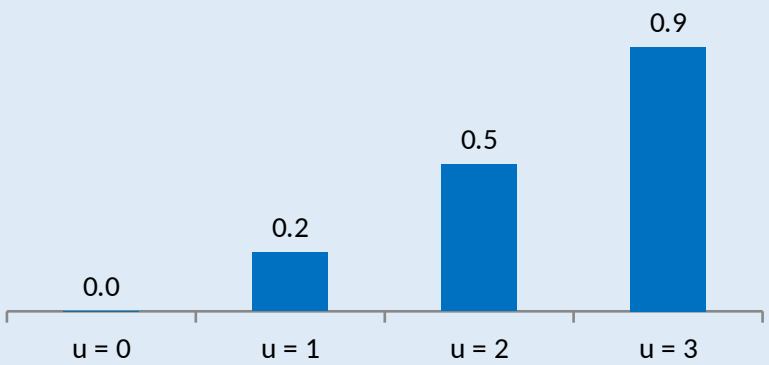
Prosody and Conversation-Stage Models

Model 2 · Urgency from prosody

Cue f_i	Factor λ_i
RMS energy	1.8
Pitch mean	1.3
Pitch variation	1.5
Speech rate	1.5
Spectral centroid	1.25

$$u = \min\left(3,\ \sum_{i=1}^{5} \mathbf{1}[f_i > \lambda_i\, b_i]\right)$$

$$r_{\text{urg}} = g(u), \qquad g:\ 0 \mapsto 0,\ 1 \mapsto 0.2,\ 2 \mapsto 0.5,\ 3 \mapsto 0.9$$



Model 4 · Conversation stage

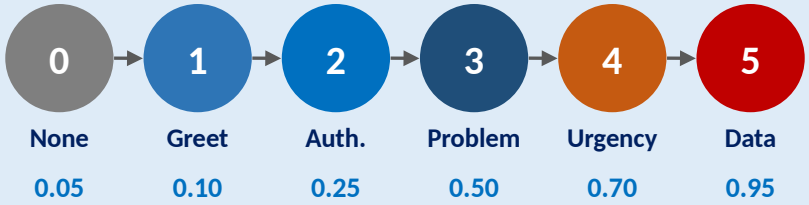
Whisper [16] speech → text → MiniLM [17], [18] text → $\mathbb{R}^{384}$ → Cosine vs intent bank

$$\cos(\mathbf{u}, \mathbf{v}) = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\|\ \|\mathbf{v}\|} = \cos\theta$$

$$s_i = \max_j\ \cos(\mathbf{e}_t, \mathbf{e}_{i,j}), \qquad i^* = \arg\max_i\ s_i$$

$$\text{stage}_t = \max\left(\text{stage}_{t-1},\ \pi(i^*)\right) \quad \text{if} \quad s_{i^*} \geq 0.55$$

otherwise unchanged: the stage only moves forward

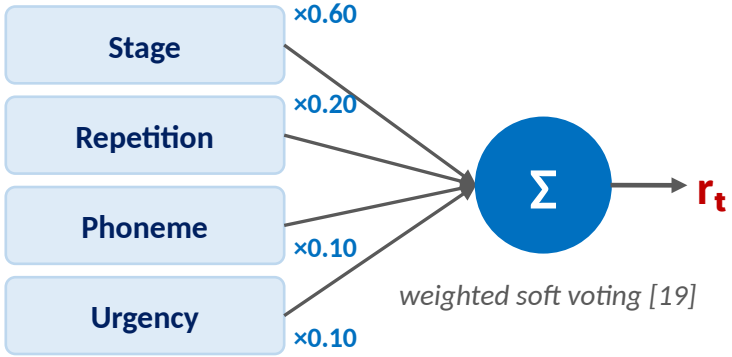


$$r_{\text{stage}} = R(\text{stage}_t)$$

5 stages × ~20 example sentences · gate 0.55

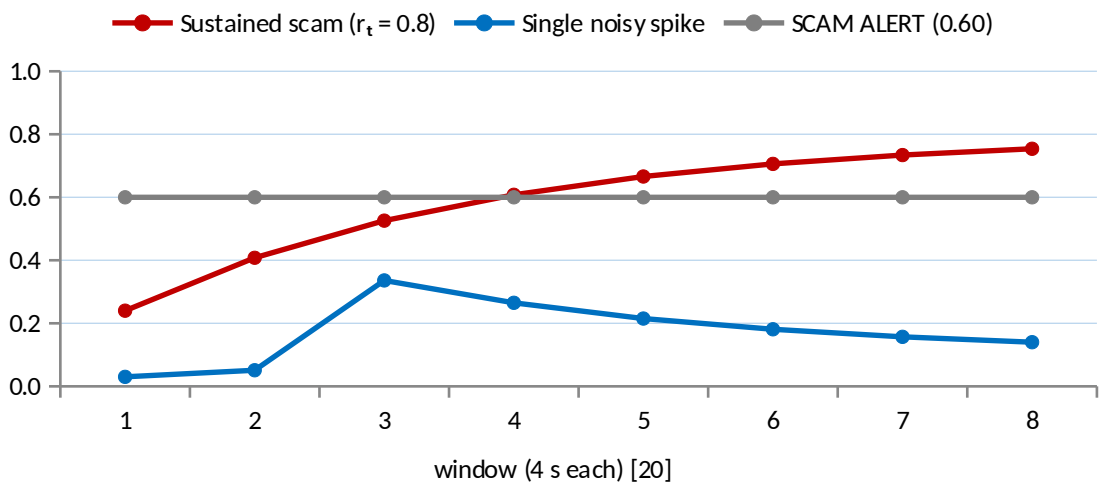
Ensemble: Weighted Fusion and EMA

$$r_t = 0.60\, r_{\text{stage}} + 0.20\, p_{\text{rep}} + 0.10\, p_{\text{phon}} + 0.10\, r_{\text{urg}}$$



Input (illustrative)	value	× w	=
Stage: PROBLEM	0.50	0.60	0.30
Repetition	0.70	0.20	0.14
Phoneme	0.60	0.10	0.06
Urgency (u = 2)	0.50	0.10	0.05
Raw risk r_t			0.55

$$R_t = \alpha R_{t-1} + \beta r_t, \qquad \alpha = 0.7,\ \beta = 0.3$$

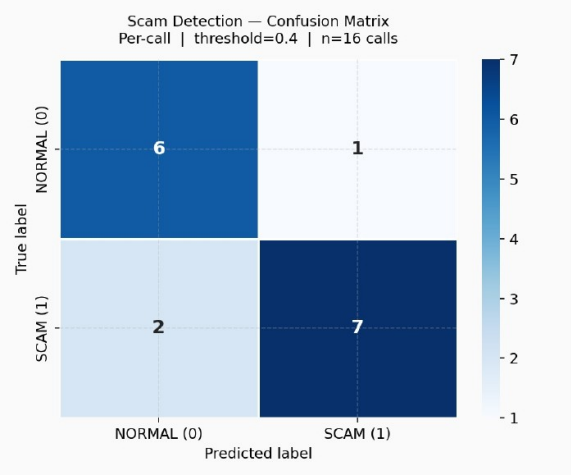


Risk label from R_t

SAFE	< 0.15
LOW	< 0.25
MODERATE	< 0.45
HIGH	< 0.60
SCAM ALERT	≥ 0.60

Results

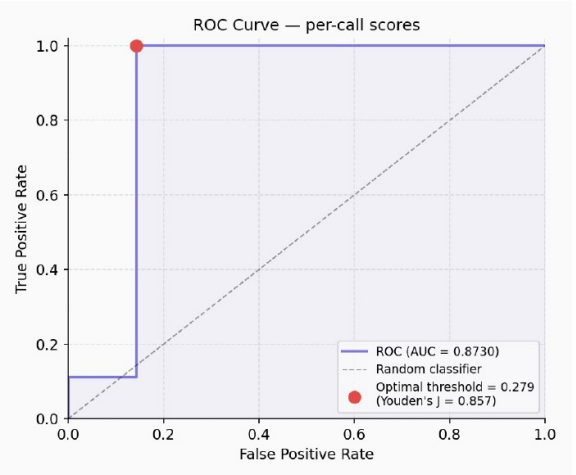
81.25%	0.813	0.873	0.875
Accuracy	Weighted F1	ROC-AUC	Scam precision

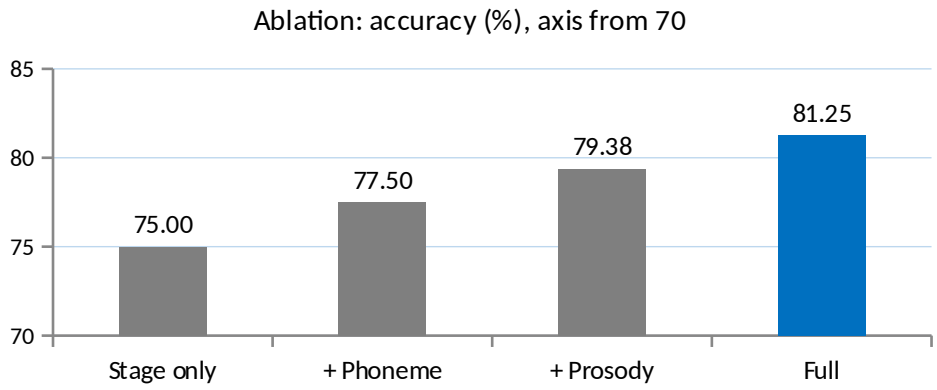


$$\text{Precision} = \frac{TP}{TP + FP} = \frac{7}{8} = 0.875$$

$$\text{Recall} = \frac{TP}{TP + FN} = \frac{7}{9} = 0.778$$

$$F_1 = \frac{2PR}{P + R} = 0.824$$





Threshold	TPR	FPR	J
0.279 (max J)	1.000	0.143	0.857
0.40 (reported)	0.778	0.143	0.635

$$J = \text{TPR} - \text{FPR}$$

16 calls (7 normal, 9 scam), live pipeline. 0.279 is tuned on these calls, so we report 0.40 [21], [22].

Discussion, Conclusion and References

Strengths

- Understands **intent**, not keywords
- Live risk **every 4 s**
- Four modalities back each other up
- EMA → few false alarms (87.5% precision)
- Complete, **open** system

Limitations → our answer

- 16 test calls → proof of concept (no public dataset)
- Whisper noise errors → 3 acoustic models + EMA
- Hand-set thresholds → explainable; learn next
- Needs internet → on-device models next

Conclusion: tracking the scam script live, fused with three acoustic cues and smoothed, gives a stable real-time warning: 81.25% accuracy, 0.873 ROC-AUC. Next: larger multilingual data, on-device inference, learned weights.

Thank you
Questions?

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